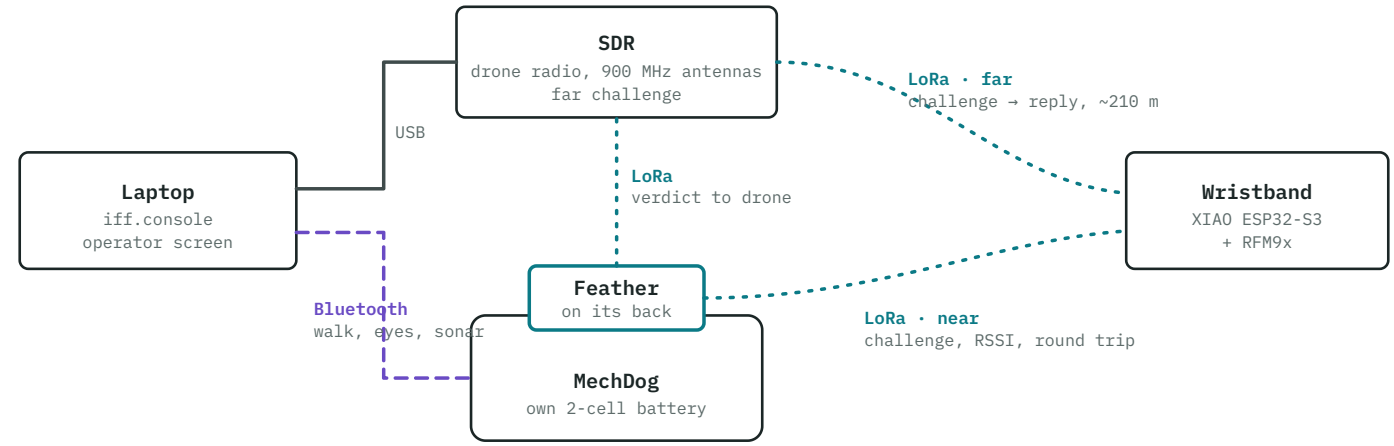
TEAM-11 · IFF BAND · THE DOG'S RADIO

MechDog LoRa Feather

An Adafruit Feather 32u4 RFM9x (900 MHz) rides on the MechDog's back as its own LoRa radio. The radio is built into the Feather, so the only wiring is the battery, a power switch and the antenna. Nothing connects to the dog itself: the laptop drives the dog over Bluetooth.

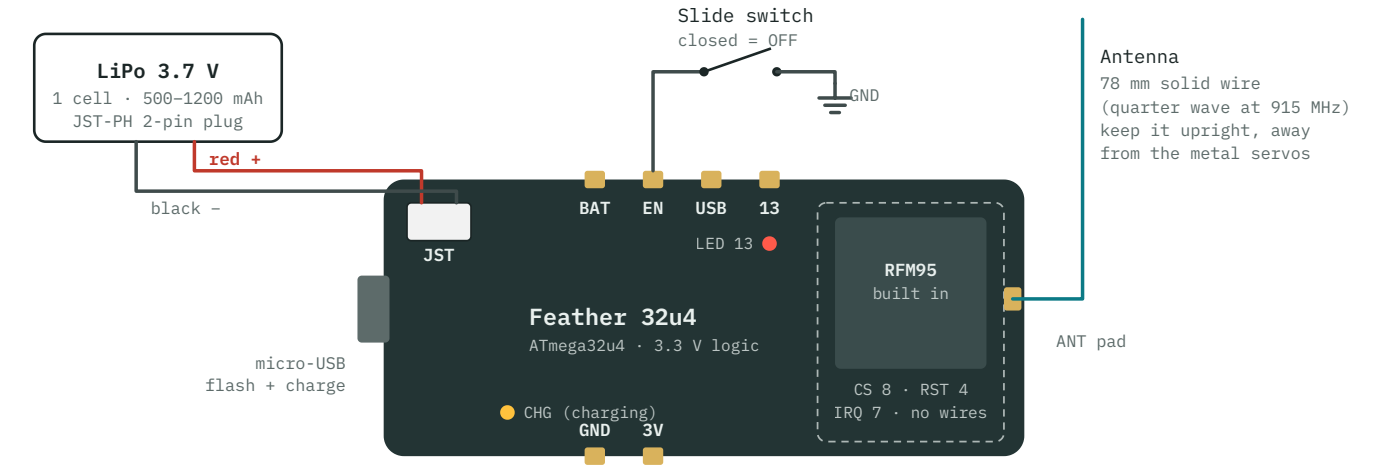
Who talks to whom



All radios: 915 MHz · SF7 · BW 500 kHz · CR 4/5 · 32-byte packets · sync word 0x12

LoRa over the air · Bluetooth · USB cable

Feather wiring



Plug USB in to flash it or charge the battery. The battery and USB can be connected at the same time. Battery voltage is readable on pin A9 through a built-in divider (×2), so the dog can report it.

battery + · ground · antenna

Connections

FROM	TO	NOTE
Power		
LiPo plug	JST socket	check red goes to +; some batteries are wired backwards
switch leg 1	**EN**	EN to GND turns the board off
switch leg 2	**GND**	still charges while off
Radio		
78 mm wire	**ANT pad**	solder before the radio ever transmits
RFM95	built in	CS 8, RST 4, IRQ 7: no wiring
The dog		
Feather	nothing	no wires to the MechDog
MechDog	laptop	Bluetooth, factory program

Don't power it from the dog's battery

The MechDog runs on a 2-cell pack, about 7.4 to 8.4 V. The Feather's battery socket takes one cell (up to 4.2 V). Plugging the dog's pack in would destroy the board. Give the Feather its own small LiPo.

Parts

- Adafruit Feather 32u4 RFM9x, 900 MHz
- 3.7 V single-cell LiPo with a JST-PH plug
- Small slide switch (SPDT or SPST)
- 78 mm of solid-core wire, 22–26 AWG
- Velcro or zip ties, a bit of foam tape
- Micro-USB data cable for flashing

Mounting on the dog

- Strap the Feather and battery to the flat top of the body, clear of the legs' swing.
- Foam tape under the board so no pad touches the metal frame.
- Antenna straight up. The wristband's antenna should point the same way.
- Leave the switch and USB port reachable.

Status

The wiring is final. The Feather firmware (the near check on board, and the report to the drone) is not written yet, and the SDR side needs GNU Radio with gr-lora_sdr. Next step: flash the Feather and check that it hears the wristband.